

In [1]: **import sys**

```
!{sys.executable} -m pip install nltk scikit-learn --break-system-pa
```

Defaulting to user installation because normal site-packages is not writeable

Requirement already satisfied: nltk in ./local/lib/python3.12/site-packages (3.9.4)

Collecting scikit-learn

Downloading scikit_learn-1.8.0-cp312-cp312-manylinux_2_27_x86_64_manylinux_2_28_x86_64.whl.metadata (11 kB)

Requirement already satisfied: click in /usr/lib/python3/dist-packages (from nltk) (8.1.6)

Requirement already satisfied: joblib in ./local/lib/python3.12/site-packages (from nltk) (1.5.3)

Requirement already satisfied: regex<=2021.8.3 in ./local/lib/python3.12/site-packages (from nltk) (2026.5.9)

Requirement already satisfied: tqdm in ./local/lib/python3.12/site-packages (from nltk) (4.67.3)

Requirement already satisfied: numpy<=1.24.1 in /usr/lib/python3/dist-packages (from scikit-learn) (1.26.4)

Requirement already satisfied: scipy<=1.10.0 in /usr/lib/python3/dist-packages (from scikit-learn) (1.11.4)

Collecting threadpoolctl>=3.2.0 (from scikit-learn)

Downloading threadpoolctl-3.6.0-py3-none-any.whl.metadata (13 kB)

Downloading scikit_learn-1.8.0-cp312-cp312-manylinux_2_27_x86_64_manylinux_2_28_x86_64.whl (8.9 MB)

8.9/8.9 MB 1.2 MB/s eta 0:00:00m eta 0:00:01[36m0:00:01

Downloading threadpoolctl-3.6.0-py3-none-any.whl (18 kB)

Installing collected packages: threadpoolctl, scikit-learn

Successfully installed scikit-learn-1.8.0 threadpoolctl-3.6.0

In [9]: **import nltk**
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk import pos_tag

In [10]: nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
nltk.download('averaged_perceptron_tagger')

[nltk_data] Downloading package punkt to /home/ubuntu/nltk_data...

[nltk_data] Package punkt is already up-to-date!

[nltk_data] Downloading package punkt_tab to /home/ubuntu/nltk_data...

[nltk_data] Package punkt_tab is already up-to-date!

[nltk_data] Downloading package stopwords to /home/ubuntu/nltk_data...

[nltk_data] Package stopwords is already up-to-date!

[nltk_data] Downloading package averaged_perceptron_tagger to /home/ubuntu/nltk_data...

[nltk_data] Package averaged_perceptron_tagger is already up-to-date!

Out[10]: True

```
In [11]: documents = [  
    "Information retrieval is the process of obtaining relevant info  
    ]
```

```
In [12]: text = documents[0]  
  
tokens = word_tokenize(text)  
  
print("Tokens:")  
print(tokens)  
  
Tokens:  
['Information', 'retrieval', 'is', 'the', 'process', 'of', 'obtaini  
ng', 'relevant', 'information', 'from', 'a', 'collection', 'of', 'd  
ocuments', '.']
```

```
In [16]: freq = nltk.FreqDist(filtered_tokens)  
  
print("Word Frequency:")  
for word, count in freq.items():  
    print(word, ":", count)
```

```
Word Frequency:  
Information : 1  
retrieval : 1  
process : 1  
obtaining : 1  
relevant : 1  
information : 1  
collection : 1  
documents : 1  
. : 1
```

```
In [ ]:
```

```
In [17]: stop_words = set(stopwords.words('english'))  
  
filtered_tokens = [w for w in tokens if w.lower() not in stop_words]  
  
print("Filtered Tokens:")  
print(filtered_tokens)
```

```
Filtered Tokens:  
['Information', 'retrieval', 'process', 'obtaining', 'relevant', 'i  
nformation', 'collection', 'documents', '.']
```

```
In [ ]:
```

```
In [15]: pos_tags = pos_tag(filtered_tokens)  
  
print("POS Tags:")  
print(pos_tags)
```

```
POS Tags:  
[('Information', 'NN'), ('retrieval', 'NN'), ('process', 'NN'), ('o  
btaining', 'VBG'), ('relevant', 'JJ'), ('information', 'NN'), ('col  
lection', 'NN'), ('documents', 'NNS'), ('.', '.')]
```

In []:

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